Tensor Networks Methods

Exercise Set 2: Entanglement & Ising spin chain

Exercise: 3-site Ising spin chain

Consider a 3-qubit spin ring (with periodic boundary conditions such that site $4 \equiv 1$) described by the Transverse-Field Ising Model (TFIM) Hamiltonian in terms of spin-1/2 operators S_i^α ($\alpha \in \{x, y, z\}$):

$$H = -J \sum_{i=1}^{3} S_i^x S_{i+1}^x - h \sum_{i=1}^{3} S_i^z = -J(S_1^x S_2^x + S_2^x S_3^x + S_3^x S_1^x) - h(S_1^z + S_2^z + S_3^z) \tag{1}$$

where $J > 0$ represents ferromagnetic exchange coupling and $h \geq 0$ is the transverse magnetic field strength in energy units. We work in the physical S^z basis $\{|\uparrow\rangle, |\downarrow\rangle\}$ with $S^z|\uparrow\rangle = +\frac{1}{2}|\uparrow\rangle$ and $S^z|\downarrow\rangle = -\frac{1}{2}|\downarrow\rangle$.

1. **Part 1: Parity Symmetry Analysis**
   (a) Define the global spin-parity operator $P = \prod_{i=1}^{3}(2S_i^z) = 8S_1^z S_2^z S_3^z$ and prove that $[H, P] = 0$.
   (b) Show that the Hilbert space splits into even and odd parts under the action of P. Write the state vectors spanning the two sectors.
2. **Part 2: Ground State Wave Function**
   (a) Show that using spatial translation symmetry, the $P = +1$ sector restricts the ground state to the 2D subspace spanned by $|v_1\rangle = |\uparrow\uparrow\uparrow\rangle$ and $|v_2\rangle = \frac{1}{\sqrt{3}}(|\uparrow\downarrow\downarrow\rangle + |\downarrow\uparrow\downarrow\rangle + |\downarrow\downarrow\uparrow\rangle)$.
   (b) Derive the ground state $|\psi_0\rangle = \cos\theta|v_1\rangle + \sin\theta|v_2\rangle$ as a function of the dimensionless field ratio $g = \frac{2h}{J}$.
   (c) Confirm that you indeed obtained the groundstate by repeating the analysis in the $P = -1$ sector.
3. **Part 3: Single-Site Reduced Density Matrix (ρ_1)**
   (a) Trace out sites 2 and 3 to compute $\rho_1 = \mathrm{Tr}_{23}(|\psi_0\rangle\langle\psi_0|)$ in the $\{|\uparrow\rangle, |\downarrow\rangle\}$ basis.
   (b) Find the eigenvalues of ρ_1 and write down the single-site von Neumann entanglement entropy $S(\rho_1)$.
4. **Part 4: Two-Site Reduced Density Matrix (ρ_{12}) & Schmidt Duality**
   (a) Trace out site 3 to derive the 4×4 density matrix $\rho_{12} = \mathrm{Tr}_3(|\psi_0\rangle\langle\psi_0|)$ in the $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$ basis.
   (b) Calculate the eigenvalues of ρ_{12} and verify Schmidt duality: $S(\rho_{12}) = S(\rho_3) = S(\rho_1)$.
5. **Part 5: Constructing the Schmidt Basis via Singular Value Decomposition (SVD)**
   (a) Express the ground state wave function $|\psi_0\rangle$ as a 2×4 coefficient matrix C between site 1 (subsystem A) and sites 2 & 3 (subsystem B).
   (b) Perform SVD on $C = U\Sigma V^\dagger$ to identify the Schmidt coefficients λ_1, λ_2 and the explicit Schmidt basis vectors $\{|u_1\rangle_1, |u_2\rangle_1\}$ and $\{|v_1\rangle_{23}, |v_2\rangle_{23}\}$.